TOPIC 04

Normal Distribution and Standardization

The bell curve, the empirical rule, z-scores, z-tables and how to compare values on different scales

WHAT YOU WILL LEARN

- Describe the normal distribution and how μ and σ shape it
- Apply the 68-95-99.7 rule
- Standardize any value with a z-score, and explain why the result has mean 0 and standard deviation 1
- Find probabilities and percentiles with a z-table or software
- Use z-scores to compare different scales and to flag outliers

1. The normal distribution

The **normal distribution** (also called the **Gaussian distribution** or the **bell curve**) is the most important distribution in statistics. We write **$X \sim N(\mu, \sigma^2)$**, meaning X is normal with mean μ and variance σ^2 (standard deviation σ).

Why it matters

- It approximates a wide range of real variables: heights and other biological measures, blood pressure, measurement error, IQ scores, many financial returns.
- **Sample means of large samples are approximately normal, whatever the population looks like** (the Central Limit Theorem, Topic 06). This is the real reason it sits at the centre of inference.
- Its mathematics is tidy: sums and averages of normal variables are normal.
- It underlies regression, confidence intervals and hypothesis tests.

The density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-(x-\mu)^2/(2\sigma^2)}$$

You never need to integrate this by hand. What matters is the shape and that probabilities are **areas** under the curve (Topic 03).

Properties

- Bell-shaped and **symmetric** about μ, so mean = median = mode.
- The curve never touches the horizontal axis, although it gets extremely close beyond about 4 standard deviations.
- Total area = 1. Exactly half the area lies on each side of μ.
- The curve changes from bending down to bending up (inflection points) at $\mu - \sigma$ and $\mu + \sigma$.
- **μ controls location** (where the centre is). **σ controls spread** (wide and flat versus tall and narrow).

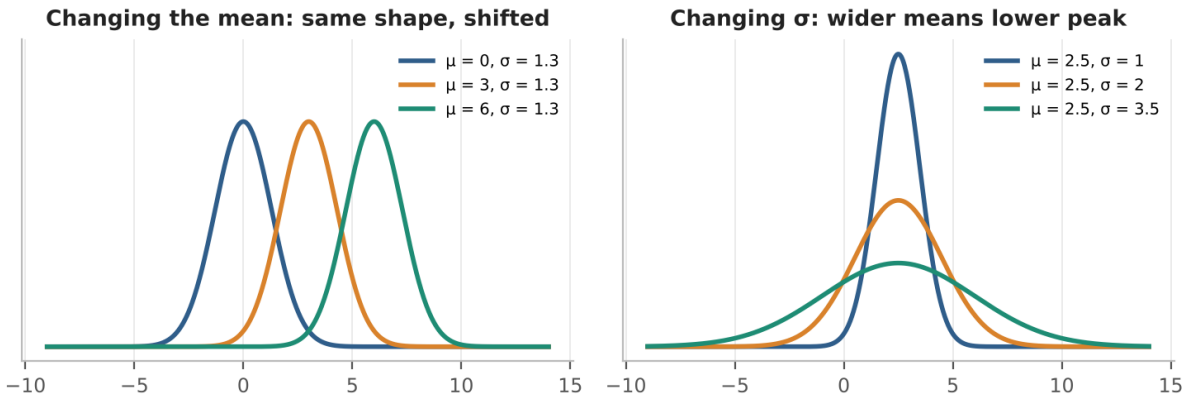


Figure 1. The mean slides the curve left or right; the standard deviation stretches or squeezes it. The area under each curve is always 1.

2. The empirical rule (68-95-99.7)

For any normal distribution the share of values within k standard deviations of the mean is fixed:

Range	Share of values	Exact
μ ± 1σ	about 68%	0.6827
μ ± 2σ	about 95%	0.9545
μ ± 3σ	about 99.7%	0.9973

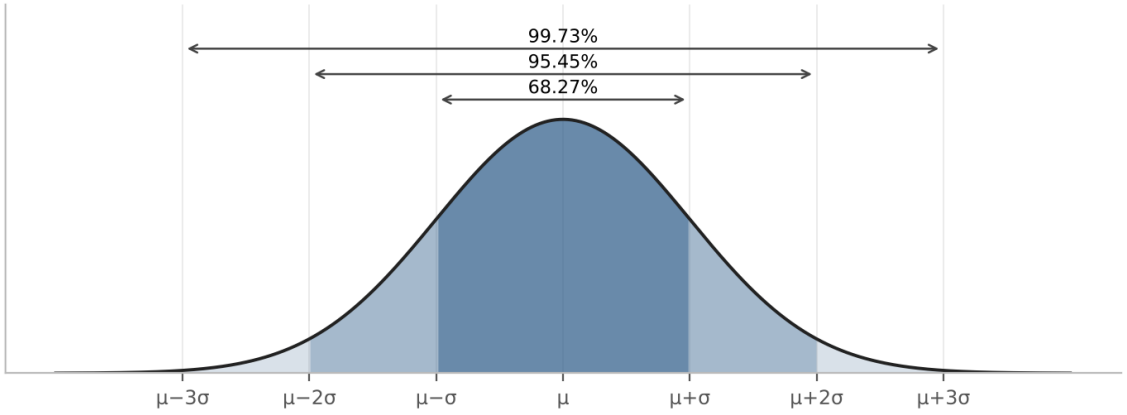


Figure 2. The 68-95-99.7 rule. Nested bands, darker means closer to the centre.

WORKED EXAMPLE **Exam marks**

Marks are roughly normal with mean 60 and σ = 8. About 68% of students score between 52 and 68; about 95% between 44 and 76; only about 0.3% fall outside 36 to 84.

3. Standardization and the z-score

There are infinitely many normal distributions (every μ and σ), but tables cannot list them all. The fix is to convert any normal variable to the one **standard normal distribution**, **Z ~ N(0, 1)**, which has mean 0 and standard deviation 1.

DEFINITION z-score

The z-score of a value x tells you **how many standard deviations x is from the mean**. The process of converting is called **standardization**. If $X \sim N(\mu, \sigma^2)$ then Z is standard normal.

$$z = \frac{x - \mu}{\sigma} \qquad \text{(sample version: } z = \frac{x - \bar{x}}{s}\text{)}$$

A z of 0 means 'exactly average', +1 means one standard deviation above, −2 means two below. The sign gives the direction; the size gives the distance in standard deviations.

Why does it work? A tiny experiment

Take the data {2, 2, 3, 3, 5}. Its mean is 3 and its standard deviation is s = 1.22. Now do the two steps of standardization separately:

Step	Data	Mean	Std dev
Original	2, 2, 3, 3, 5	3.00	1.22
1. Subtract the mean	−1, −1, 0, 0, 2	0.00	1.22
2. Divide by the std dev	-0.82, -0.82, 0.00, 0.00, 1.63	0.00	1.00

- **Subtracting a constant** slides the data; it moves the mean (to 0) but leaves the spread unchanged.
- **Dividing by a constant** rescales the data; it divides the spread by that constant, so dividing by s makes the standard deviation exactly 1.
- Standardization changes the *units* and *location*, never the *shape* of the distribution.

4. From z to probability: the z-table

The **standard normal table** lists the **cumulative probability** $P(Z \leq z)$, the area to the left of z. To read it, split z into its first decimal (row) and second decimal (column). For z = 1.96, go to row 1.9 and column 0.06 to find **0.9750**.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952

Excerpt of the standard normal table: P(Z ≤ z). The complete table is in the course Excel file.

Four patterns cover every question

Question	How to compute	Example (z = 1.25)
$P(Z \leq z)$	Read the table directly	0.8944
$P(Z > z)$	$1 - P(Z \leq z)$	0.1056
$P(a < Z < b)$	$P(Z \leq b) - P(Z \leq a)$	$P(-1 < Z < 1.25) = 0.7357$
Negative z	Use symmetry: $P(Z \leq -z) = P(Z > z)$	$P(Z \leq -1.25) = 0.1056$

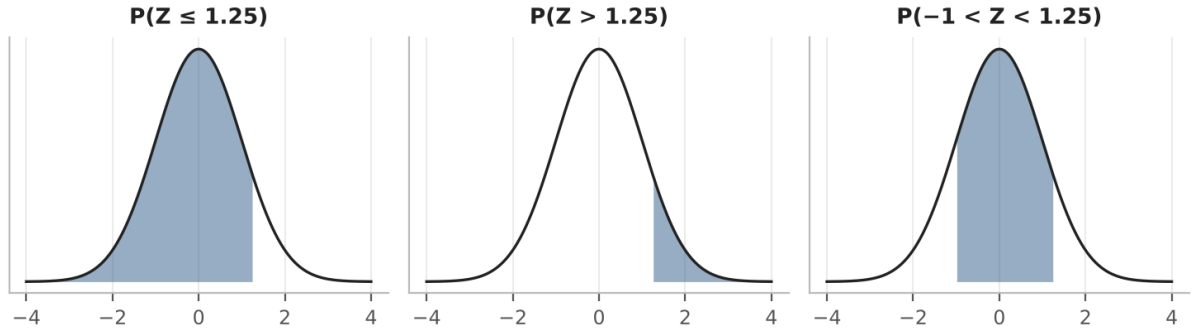


Figure 3. The shaded area is the probability. Sketch the curve and shade before you calculate.

COMMON MISTAKE **Sketch first**

Most z-table errors come from using the wrong tail. The table gives the area to the **left**. Draw the bell, shade what the question asks for, and decide whether to subtract from 1.

5. Worked examples

WORKED EXAMPLE **1. A probability from raw values**

IQ scores follow $N(100, 15^2)$. Find $P(IQ > 130)$. $z = (130 - 100)/15 = 2.00$. $P(Z > 2) = 1 - 0.9772 =$ **0.0228**, about 2.3% of people.

WORKED EXAMPLE **2. A probability between two values**

Same IQ distribution: $P(85 < IQ < 115)$. z-values are -1 and $+1$, so the probability is $0.8413 - 0.1587 =$ **0.6827** (the 68% of the empirical rule).

WORKED EXAMPLE **3. Working backwards: a percentile**

What IQ separates the top 10%? We need the 90th percentile: z with $P(Z \leq z) = 0.90$ is $z = 1.2816$. Convert back with $x = \mu + z\sigma = 100 + 1.2816 \times 15 =$ **119.2**.

WORKED EXAMPLE **4. Comparing different scales**

Ana scores 1300 on a test with mean 1050 and $\sigma = 200$. Ben scores 28 on another test with mean 21 and $\sigma = 5$. Who did better relative to their group? Ana: $z = (1300 - 1050)/200 = 1.25$. Ben: $z = (28 - 21)/5 = 1.40$. **Ben** did better: 1.40 standard deviations above average versus 1.25. Percentiles: Ana 89.4th, Ben 91.9th.

Critical z-values to remember

These values cut off a chosen tail probability and appear constantly in confidence intervals and tests:

Confidence level	α	Two-tailed critical value z*	One-tailed critical value
90%	0.10	1.645	1.282
95%	0.05	1.960	1.645
99%	0.01	2.576	2.326

6. Using z-scores in practice

Spotting outliers

Under normality only 0.27% of values have $|z| > 3$, so such points are unusual. A common screening rule flags $|z| > 3$ (sometimes 2.5). Remember that outliers inflate $\bar{x}$ and s, and so can hide themselves; check them rather than deleting automatically.

Is my data actually normal?

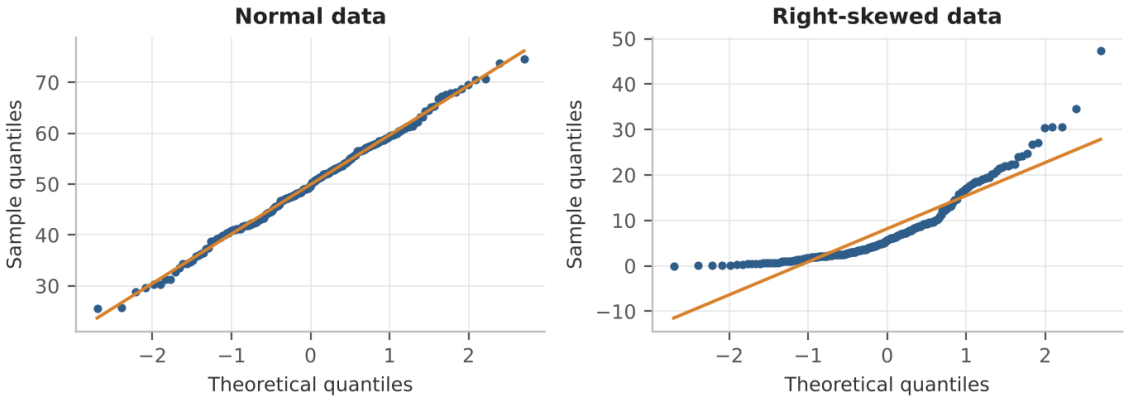


Figure 4. Q-Q plots. Points hugging the line indicate normality; a curved pattern (right) signals skew.

Check normality with a histogram, a Q-Q plot like Figure 4, or a formal test (Shapiro-Wilk). With large samples the Central Limit Theorem often makes exact normality unnecessary for inference about means.

COMMON MISTAKE

- Using a z-table when σ is unknown and the sample is small; that calls for the t distribution (Topic 09).
- Forgetting that the table gives the **left** tail.
- Treating a z-score as a probability. A z of 2 means 2 standard deviations away, not 2%.
- Assuming 'standardized' means 'normal'. Standardizing a skewed variable leaves it skewed.

7. In Python

```
from scipy import stats
import numpy as np

stats.norm.cdf(1.96)                        # P(Z <= 1.96)         0.9750
stats.norm.sf(2)                            # P(Z > 2)             0.0228
stats.norm.ppf(0.975)                       # z for 97.5th pct     1.9600
stats.norm(100, 15).cdf(130)                # P(IQ <= 130) without standardizing by hand
stats.norm(100, 15).ppf(0.90)               # 90th percentile of IQ -> 119.2

x = np.array([2, 2, 3, 3, 5])
z = (x - x.mean()) / x.std(ddof=1)          # or scipy.stats.zscore(x, ddof=1)
```

SUMMARY

- $X \sim N(\mu, \sigma^2)$: symmetric bell; μ sets location, σ sets spread; probability is area.
- 68-95-99.7: share of values within 1, 2, 3 standard deviations.
- $z = (x - \mu)/\sigma$ counts standard deviations from the mean; $Z \sim N(0, 1)$.
- Subtracting μ makes the mean 0; dividing by σ makes the standard deviation 1.
- Read $P(Z \leq z)$ from the table; use $1 -$ for right tails and symmetry for negative z.
- Key values: 1.645, 1.96, 2.576 for 90%, 95%, 99% two-sided.

Practice questions

Try each question before reading the answer underneath it.

Q1. Heights are $N(170, 8^2)$ cm. What percentage of people are taller than 186 cm?

Answer. $z = 2.00$, so $P(Z > 2) = 0.0228$, about 2.3%.

Q2. Find $P(Z$ between -1.5 and $0.5)$.

Answer. $0.6915 - 0.0668 = 0.6247$.

Q3. Battery life $\sim N(500, 40^2)$ hours. Find the 5th percentile.

Answer. $z = -1.645$; $x = 500 + (-1.645)(40) = 434.2$ hours.

Q4. A value has $z = -0.8$. Interpret it.

Answer. It is 0.8 standard deviations below the mean, at about the 21th percentile.

Q5. A dataset has mean 12 and standard deviation 4. After subtracting 12 and dividing by 4, what are the new mean and standard deviation?

Answer. Mean 0, standard deviation 1.

Q6. Why can you not use the empirical rule on a strongly skewed variable?

Answer. The 68-95-99.7 percentages are properties of the normal distribution. A skewed distribution has a different shape, so those shares do not hold.